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Task Description:	We have identified recessive mutations at three loci defective in the signal transduction events of gravitropism in Arabidopsis. The gravity persistent signal (gps) mutants were isolated using a temperature effect on gravity signal transduction and response. gps1 did not bend in response to the 4 degrees C gravity stimulation upon return to 21 degrees C. gps 2 responded to the 4 degrees C gravity stimulus with the same kinetics, but in the opposite direction as wild type. gps 3 mutants over-responded after return to room temperature, continuing to bend past the predicted 90 degrees reorientation. The gps mutants are some of the first mutants to be reported that are affected in aspects of gravitropic signal transduction prior to auxin transport, including how polarity is established. We have cloned the genes defective in all three mutants and are now characterizing their expression in wild-type plants so that we can better understand their role in gravitropism. In addition, we have identified several potential new mutants in the GPS mechanism.		

	The objective of the proposed research was to identify components of the gravity signal transduction pathway. We have identified a group of mutants specifically defective in the events of signal transduction and the specific objectives of this research were to characterize the mutants, clone the genes affected, and identify new mutants in the pathway. The research has provided us with more data on the molecular components of gravity signal transduction, an area that remains a “black box” in plant gravitropism. This data collected has provided further insight into how plant growth and development is affected by gravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Gravitropism is an important developmental process that allows plant organs to use gravity as a guide for growth. In inflorescence stems, gravity sensing seems to involve the sedimentation of amyloplasts in the endodermis. Following the displacement of amyloplasts upon gravistimulation, lateral redistribution of auxin is proposed to initiate differential growth along the elongation zone that results in curvature. The primary objective of this proposal is to improve our understanding of the signal transmission phase of gravitropism through the study of a new group of mutants defective in the early signal transduction events that link amyloplast sedimentation to auxin redistribution. The work completed has provided data allowing a better understanding of the signal transduction events that establish auxin redistribution during gravitropism. The identification of genes involved in the GPS response should enhance our knowledge of the components of the signal transduction pathway leading from gravity stimulus perception to the differential growth response of inflorescence stems. To date, little is known about this important aspect of plant gravitational biology. Gravitropism allows plants to orient their organs optimally for photosynthesis or water and nutrient uptake. Besides providing new, fundamental information to supplement existing models of plant gravitropism, elucidating factors governing plant responses to gravitational forces has potentially far-reaching implications for manipulating plant growth and development for human use (e.g. advances in applied research that might enable the manipulation of plant organ position, increasing access for harvest, and lead to innovations that will improve plant use for bioregenerative life support for extended spacefight.) and for understanding other signal transduction events, especially those initiated by biophysical stimuli. In addition, polar auxin transport controls multiple aspects of plant growth and development, including tropisms, embryo axis formation, vascular development, lateral root formation and apical dominance. Therefore, a better understanding of the events that establish auxin redistribution during gravitropism could increase our fundamental understanding of the processes involved in plant development.
Task Progress:	The proposed research focused on the genetic and physiological characterization of the gravity persistent signal (gps) mutants of Arabidopsis. The gps mutants were isolated using a chilling effect on gravity signal transduction that separated stimulus perception from response. gps1 mutants did not bend in response to the 4 degrees C gravity stimulation upon return to 21 degrees C. gps 2 responded to the 4 degrees C gravity stimulus by bending in the opposite direction to wild type. gps 3 mutants over-responded after return to room temperature. In the first year of the project, we were able to characterize all three mutants with regard to their physiology, statolith sedimentation and auxin transport. We were also able to clone the genes defective in gps1 and gps3. In the second year, we performed subcellular localization of these genes, made strides to complement the mutant phenotype and made the necessary constructs toward assessing expression of the gene in vivo using a green fluorescent protein fusion. We have also performed a transcription profiling experiment with gps1 in hope of better understanding the effect of the mutation of expression of other genes involved in gravitropism and have identified a family member of GPS1 that is expressed in root tissue. In addition, we have cloned the gene defective in gps2 and have identified several new potential mutants in the GPS pathway. In year 3, we finalized assessment of auxin redistribution in all the mutants and have further characterized the root-specific GPS1 family member and have made significant progress in characterizing the protein. We have also used several bioinformatics tools to gain more information on all three of the GPS genes, have finalized all constructs needed for assessment of temporal and spatial expression and transformed them into their respective mutant background, and have used qRT-PCR to evaluation gene expression patterns. GPS1 was cloned using Thermal Asymmetric Interlaced Polymerase Chain Reaction (TAIL PCR). The gene identified in both gps1-1 and gps1-4 was CYP705A22, a p450 of unknown function, located on chromosome 3 of Arabidopsis. Building constructs for complementation of the gps1 phenotype has proven problematic, but we now have a construct for the genomic DNA for CYP705A22 under control of both the 35S promoter and the endogenous promoter ready for transformation into gps1 plants. To characterize timing and tissue specificity of gene expression, we have also constructed a variety of green fluorescent protein (GFP) fusions. We have transiently expressed a GFP:CYP705A22 transit peptide fusion in onion cells and BY2 tobacco suspension culture cells providing information on subcellular localization. The gene appears to be targeted to the endomembrane system. In addition, the promoter for the CYP705A22 has been fused to GFP to test for tissue specificity and timing of expression. Analysis of these fusion proteins will better our understanding of how this gene/protein may be involved in gravitropic signal transduction. In addition, through analysis of microarray data generated from gravistimulated root tips, we were able to identify a root specific P450 in the same family as GPS1. This may indicate that the P450 family members are not only specific for gravity but tissue specific as well. The biological activity of CYP705A5, like CYP705A22, is unknown. To further characterize these P450s, we obtained several Arabidopsis lines containing mutations in the root specific P450. Analysis of the gravitropic response of these mutants indicates that root gravitropism was significantly reduced in the mutant, although the growth rate is unaffected. Using the phylogenetic relationship among the P450s, we have determined that CYP705A22 (and CYP705A5) are most similar in structure to CYP93C. The CYP93C family of P450s is involved in flavonoid biosynthesis in Lotus japonicus, specifically the biosynthesis of the isoflavan phytoalexins, vestitol and sativan. This result has led to the hypothesis that CYP705A22 and A5 may be involved in flavonoid biosynthesis. The hypothesis is supported by the increased polar auxin transport in gps1 as in tt4. However, having the root mutants has allowed us to chemically complement the root phenotype. Addition of dihydroquercetin, but not narigenin, to the growth media restored (enhanced) the root curvature to that of wild-type, further supporting our hypothesis that CYP705A22 and A5 may be involved in flavonoid biosynthesis. (A P450 is one intermediate in the conversion of naringenin to dihydroquercetin in the phenylpropanoid/flavonoid biosynthetic pathway.) To determine substrate binding for the P450s involved in this proposal, we have established a collaboration with Dr. Mary Schuler at the University of Illinois. We also performed a transcription profiling experiment with gps1. We analyzed RNA extracted from wild-type and gps1

plants 5 minutes after cold gravistimulation and after cold treatment alone. Analysis of these data gave us insight into additional genes that were indirectly affected by the gps1 mutation. We identified 44 genes specific to the gravity stimulus that were not expressed in the cold treatment alone and 76 that were differentially expressed between wild type and gps1 after cold gravistimulation. Of these, we have selected seven and ten genes, respectively, based on the level of expression and analyzed their expression. Those genes showing highest fold increase in gps1 when compared to wild-type after GPS treatment included an F-box protein, zinc finger family protein, bZIP transcription factor and putative phospholipase. Also, the finding that gps1 mutant is defective in a cytochrome P450 like gene (preliminary data) could help us to investigate the role of cytochrome P450's on these genes. Spot 3 vacuolar protein, methyl crotonyl CoA carboxylase and the hypothetical proteins show a pattern of increase 30min after gravistimulation which is lacking in gps1 mutant which suggests that these events might be down stream of the GPS1 gene. These data can be used to potentially design more experiments to better understand the gravity signal transduction. The T-DNA insertion in gps3 disrupts a non-coding region on chromosome 1. The region is in the 3' untranslated region between two genes, an auxin response factor 9 (ARF9) – like gene and a hypothetical gene. We have now determined that the ARF9 like gene is actually the gene disrupted in gps3. We have used a genomic clone of the gene with its endogenous promoter to complement function in the mutant, and using transient expression of a GPS3::GFP fusion have confirmed a nuclear localization of the gene product. We have also built fusions between GFP and GPS3 under control of the endogenous promoter. Those constructs have been transformed into plants and selection of transformants is underway. These plants will be used to determine tissue specificity and temporal expression of the protein.

The gene defective in gps2 has now been cloned as well. GPS2 is At5g11150 and encodes a synaptobrevin/ vesicle associated protein that is postulated to be involved in vesicle trafficking. All additional alleles show a defect in the same gene. Constructs have been made to complement the mutant and do temporal and spatial assessment of gene expression and analysis of plants has begun.

And finally, through a continuation of the gps mutant screen, we have identified 12 putative new gps mutants to date and the screen continues. A continuation of the gps mutant screen has identified 12 putative new gps mutants. The new mutants fall into the six phenotypic groups: two already identified: wrong way and no response, and four new phenotypes: delayed responder, rapid responder, random responder and the under-achiever. Once complementation groups are being determined, new mutants that do not fall into the complementation groups previously identified will be selected for molecular characterization. Based on these data, we will select 1-2 of the most promising new mutants for continued characterization.

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Presentation	McCallister, J.; King,R.; Wyatt, S.E. "Molecular characterization of the gravity persistence signal (gps) 2 mutant in Arabidopsis thaliana" Gordon Conference on Mechanotransduction and Gravity Signaling (Biddeford ME : July 2005) Jul-2005
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